The best calculations we have suggest that our nice pleasant vacuum is not likely to undergo a radical rearrangement anytime soon—at the time of writing, the latest estimate gives us more than 10^{100} years. By then, we'll likely be well into the process of a Heat Death, or perhaps, if we're very unlucky, being torn asunder by a Big Rip. At which point maybe an instant painless obliteration won't seem so bad.

So, technically, I can't tell you for certain that vacuum decay isn't right about to happen. I also can't tell you for certain that it hasn't already happened somewhere in our own Solar System, or on the other side of the galaxy, or in another galaxy, creating a light-speed-expanding bubble that is silently coming for us as we speak. But I can tell you that if you'd like to prioritize your paranoia, you're *much* more likely in your lifetime to be struck by lightning, an out-of-control car, rampaging cattle, or even a stray meteor than by the spontaneous appearance of a bubble of true vacuum.

There's just one more thing, though.

We've already covered the facts that we can't produce our own vacuum decay bubble with a high-energy particle collision, and that a spontaneous tunneling event is so unlikely we should probably try very hard to forget that we ever heard of it in the first place. But recently, physicists have come up with yet another way to destroy the universe with vacuum decay, and I have to say it's kind of a cool one.

SMALL BUT DEADLY

In 2014, Ruth Gregory, Ian Moss, and Benjamin Withers, building on a bit of previous work on this topic, put out a new paper that caught my attention. It explained that while spontaneous vacuum decay is tediously slow, the presence of a black hole could speed the process up considerably and generally make things much more interesting. In fact, they argued, the really dangerous thing is a *small* black hole, because particle-sized black holes can dramatically increase the chance of vacuum decay occurring right on top of them. Maybe we don't have to wait 10^{100} years after all.

The way it works is similar to the way a particle of dust can condense a bit of water around it in a humid room, or the way that clouds get seeded in the upper atmosphere. The dust particle is a *nucleation site*—something that differentiates that point from others and allows the process to happen more easily. In the case of clouds and water, the water molecules have an easier